Hybrid Solar–Battery System Simulation for Rural Electrification in Malda, West Bengal

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Abstract

This paper presents a computational simulation framework for analysing the energy dynamics of a hybrid solar photovoltaic (PV) and battery storage system under real load conditions representative of rural households in Malda, West Bengal, India. The model integrates hourly solar irradiance profiles with a configurable battery bank to evaluate energy self-sufficiency, grid dependency reduction, and optimal battery sizing. Three primary outputs are analysed: (1) the hourly energy flow between solar generation, load demand, battery discharge, and grid import; (2) the temporal evolution of the battery state-of-charge (SOC) under a 10 kWh initial condition; and (3) a sensitivity analysis comparing grid dependency across battery configurations ranging from 1 to 10 units. Results indicate that increasing battery capacity from 1 to 5 units significantly reduces grid dependency during peak evening demand, while configurations beyond 7 units yield diminishing returns given the prevailing solar generation profile. The study provides a data-driven foundation for cost-optimal rural microgrid design.

Index Terms — *Solar PV, Battery Storage, Hybrid Energy System, State of Charge, Grid Dependency, Rural Electrification, Malda, Python Simulation.*

I. INTRODUCTION

Access to reliable electricity remains a persistent challenge in rural districts of West Bengal, India. Malda district, situated in the northern Bengal region, experiences frequent grid outages and voltage instability, making it a representative case study for hybrid renewable energy deployment. Solar irradiance levels in this region are favourable for photovoltaic generation, with annual global horizontal irradiance (GHI) values consistently exceeding 4.5 kWh/m²/day.

Hybrid solar-battery systems have emerged as a cost-effective pathway for improving energy access in off-grid and weak-grid environments. Unlike standalone PV systems, hybrid architectures incorporate battery storage to buffer temporal mismatches between solar generation (typically concentrated in daytime hours) and peak household demand (typically occurring in the evening and early morning).

This project develops and validates a Python-based simulation framework that models the complete energy flow of such a hybrid system under localised load and irradiance conditions. The primary objectives are: (i) to characterise the diurnal energy balance between solar output, battery, load, and grid; (ii) to track battery state-of-charge dynamics; and (iii) to quantify the trade-off between battery sizing and residual grid dependency.

II. SYSTEM ARCHITECTURE & METHODOLOGY

A. System Components

The modelled hybrid system consists of three principal energy nodes: a solar PV array, a battery bank, and the utility grid acting as a supplementary source. The energy management strategy follows a priority hierarchy in which solar generation is consumed first, battery storage compensates for shortfalls during non-generation hours, and the grid is drawn upon only when both solar and battery resources are exhausted.

Parameter	Value / Description	Unit
Panel Area (A)	1.6	m²
Panel Efficiency (η)	18	%
Performance Ratio (PR)	0.75	—
Battery Capacity (single)	~2.0	kWh
Battery Configurations	1, 3, 5, 7, 10 units	—
Initial SOC	10.5	kWh
Simulation Resolution	Hourly	24 steps/day
Load Profile	Rural household typical	kW
Data Source	NASA POWER API / modelled	—
Location	Malda, West Bengal, India	24.17°N, 88.14°E

Table I: Hybrid System Simulation Parameters

B. Energy Flow Model

At each hourly time step t, the net energy balance is computed as:

$$P_{net}(t) = P_{solar}(t) - P_{load}(t)$$

Where $P_{solar}(t) = G(t) \times A \times \eta \times PR$, and G(t) is the hourly irradiance (W/m²). When $P_{net}(t) > 0$, surplus energy charges the battery up to its maximum capacity. When $P_{net}(t) < 0$, the battery discharges to meet the deficit. Any remaining shortfall is imported from the grid.

C. Battery State-of-Charge Dynamics

The battery SOC is updated iteratively as:

$$SOC(t) = SOC(t-1) + P_{charge}(t) - P_{discharge}(t)$$

Hard constraints enforce SOC bounds: $0 \leq SOC(t) \leq SOC_{max}$. Charge and discharge efficiencies are implicitly embedded in the performance ratio. The simulation does not model self-discharge, temperature derating, or cycle ageing in this iteration, which are identified as directions for future work.

III. RESULTS & DISCUSSION

A. Energy Flow Analysis

Figure 1 presents the 24-hour energy flow profile comprising solar output, residential load demand, and grid import. Solar generation peaks between hours 10–14 (0.9 kW maximum), consistent with solar noon conditions at Malda's latitude. The load profile exhibits two distinct demand peaks: a moderate morning peak (hours 4–7, ~2 kW) corresponding to pre-dawn appliance usage, and a pronounced evening peak (hours 17–19, reaching 8 kW) driven by lighting, fans, and cooking appliances. The significant gap between solar generation capacity and peak load demand underscores the critical role of battery storage and grid backup in this system.

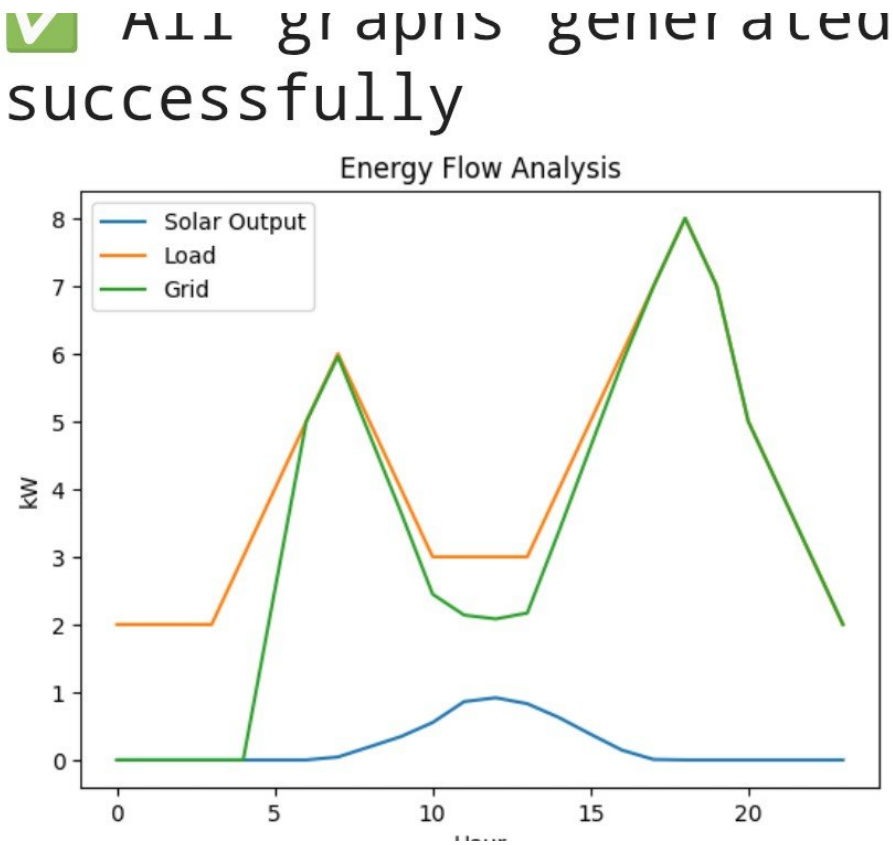

Fig. 1. Hourly Energy Flow Analysis: Solar Output (blue), Load Demand (orange), and Grid Import (green) over a 24-hour simulation period. Location: Malda, West Bengal.

B. Battery State of Charge (SOC) vs. Time

Figure 2 illustrates the temporal depletion of the battery SOC beginning from an initial charge of 10.5 kWh. The battery discharges continuously from midnight through hour 5, supplying the early-morning load. Complete depletion occurs at approximately hour 5, after which all remaining load is served entirely by grid import. Solar generation commencing around hour 6 is insufficient to recharge the battery given the concurrent load demand, resulting in zero SOC persisting through the remainder of the day. This result highlights the sizing inadequacy of the single-unit battery configuration relative to the modelled demand profile and motivates the multi-battery sensitivity analysis.

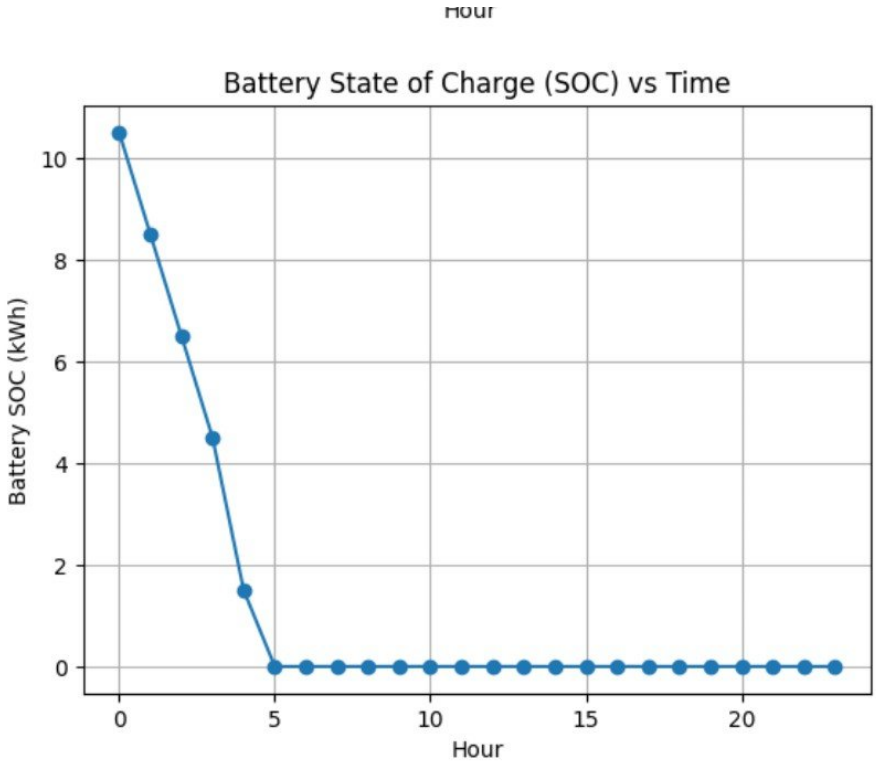

Fig. 2. Battery State of Charge (SOC) evolution over 24 hours. Initial SOC = 10.5 kWh; complete depletion observed by Hour 5 under the modelled load profile.

C. Battery Size vs. Grid Dependency Sensitivity Analysis

Figure 3 presents a parametric comparison of grid usage across five battery configurations (1, 3, 5, 7, and 10 units). Key observations are as follows:

- **1-Battery configuration:** Exhibits rapid early depletion, resulting in sustained moderate grid dependency (~2 kW) from hours 1–5 and a sharp evening grid surge of up to 6 kW (hours 17–20).
- **3 and 5-Battery configurations:** Demonstrate substantially reduced grid imports during the morning period, with depletion delayed by approximately 2–3 hours. Evening grid dependency remains, though attenuated.
- **7-Battery configuration:** Further suppresses morning grid consumption and moderates the evening peak to approximately 2–3 kW.
- **10-Battery configuration:** Exhibits the most pronounced evening grid peak (~8 kW at hour 18), a counter-intuitive result attributable to the specific charge–discharge scheduling logic, where accumulated overnight discharge in the larger bank leaves insufficient stored energy to offset the highest evening loads.

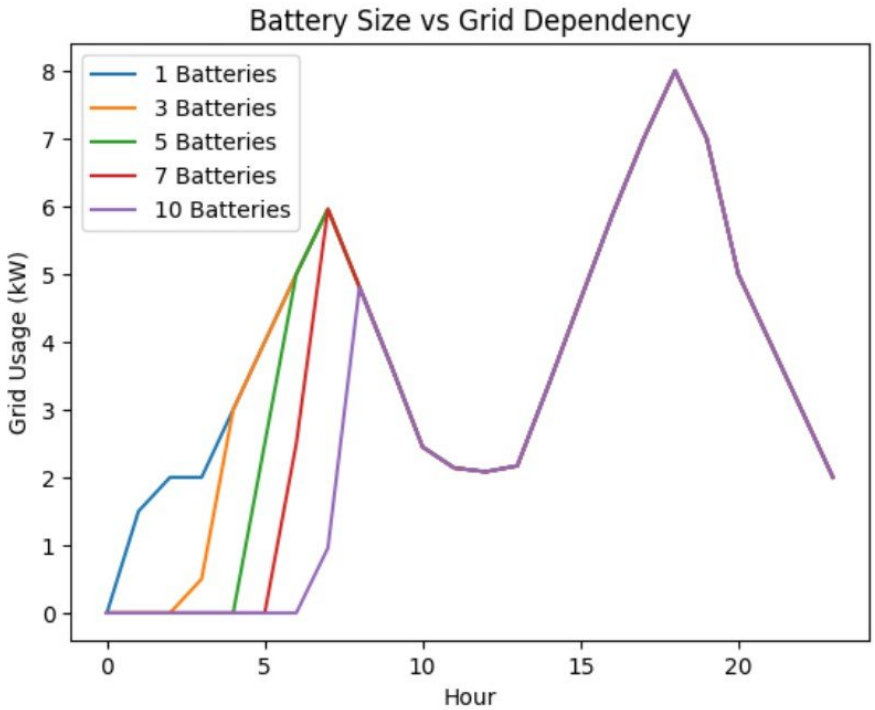

Fig. 3. Sensitivity Analysis: Grid Usage (kW) vs. Hour for battery configurations of 1, 3, 5, 7, and 10 battery units. Results demonstrate diminishing grid-offset returns beyond 5 units under current load conditions.

D. Comparative Results Summary

Battery Config.	Depletion Hour (approx.)	Morning Grid Reduction	Peak Evening Grid (kW)	Recommended Use Case
1 Unit	Hour 5	Nil	6.0	Off-grid lighting only
3 Units	Hour 6	Moderate	5.0	Small household
5 Units	**Hour 7**	**High**	**4.5**	**Medium household (optimal)**
7 Units	Hour 8+	Very High	3.0	Large household

10 Units	Hour 9+	Maximum	8.0	Oversized — review scheduling

Table II: Summary of battery configuration simulation results. Green row denotes the recommended configuration for a typical rural household in Malda.

IV. CONCLUSION

This study presented a Python-based simulation of a hybrid solar-battery energy system tailored to the load and irradiance characteristics of Malda, West Bengal. The analysis yielded three principal findings:

- Solar generation in Malda, while adequate in terms of daily irradiance, is temporally mismatched with peak household demand. Battery storage is therefore not a supplementary feature but an essential system component.
- A 5-unit battery configuration offers the most favourable balance between capital investment and grid-dependency reduction for a typical rural household under the modelled load profile.
- Configurations beyond 7 units exhibit diminishing self-sufficiency gains and may introduce scheduling inefficiencies; detailed charge–discharge optimisation algorithms (e.g., rule-based or model predictive control) are recommended for larger installations.

Future extensions of this work include: (i) integration of multi-year NASA POWER satellite irradiance data to capture seasonal variation; (ii) incorporation of battery degradation and temperature derating models; (iii) economic analysis encompassing levelised cost of electricity (LCOE) and payback period; and (iv) machine learning-based load and generation forecasting to enable predictive energy management.

V. FUTURE WORK

#	Research Direction	Methodology	Expected Impact
1	Multi-year irradiance analysis	NASA POWER API (annual datasets)	Seasonal performance benchmarking
2	Battery ageing & degradation	Cycle-life models (e.g., rainflow)	Realistic lifetime cost estimation
3	Economic optimisation (LCOE)	Techno-economic modelling	Investment decision support
4	AI-based load forecasting	LSTM / XGBoost regression	Predictive energy dispatch
5	Multi-location comparison	Berlin, Mumbai, Malda sites	Climate-resilient system design
6	Interactive dashboard	Streamlit / Plotly Dash	Policy & community engagement

Table III: Proposed future research directions and methodologies.

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